

PHASE 1 PROJECT REPORT

AUTOMATIC TOUCHLESS WATER DISPENSER

Arduino-Based Touchless Dispensing Using Ultrasonic Detection, I2C LCD and Dual Servo Control

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Phase 1: Electronic Design, Embedded Programming and 3D Enclosure Development

1. Abstract

The Automatic Touchless Water Dispenser is an embedded-system project designed to operate a water tap without requiring direct physical contact. An HC-SR04 ultrasonic sensor measures the distance of a user's hand. When a valid object is detected within 15 cm, an Arduino Uno activates two servo motors to operate the dispensing mechanism. A 16×2 I2C LCD provides user messages and distance information. Phase 1 focuses on circuit design, embedded programming, system integration planning and development of a 3D-printable enclosure. The design also provides a base for future water-level sensing, rechargeable power, ESP32 IoT monitoring and usage analytics.

2. Introduction and Problem Statement

Manually operated water taps require physical contact and can remain open longer than necessary. A touchless mechanism can improve convenience and reduce repeated contact with the tap. This project combines non-contact ultrasonic sensing, Arduino control, servo actuation, LCD feedback and a custom enclosure to automate the basic dispensing sequence.

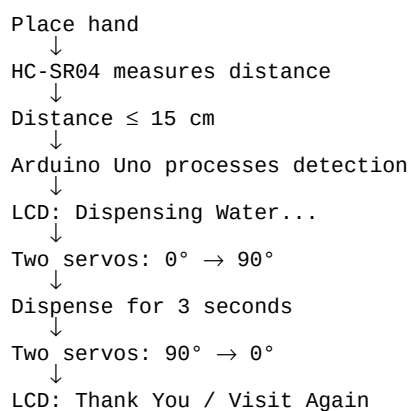
3. Objectives

The objectives are to detect a user's hand automatically, provide touchless tap actuation, display system status on an LCD, implement repeatable servo opening and closing, develop the electronic circuit and Arduino software, and design 3D-printable enclosures for the electronics and mechanical actuator.

4. Working Principle

The HC-SR04 continuously measures distance. The Arduino Uno compares the measurement with a 15 cm threshold. When a hand is detected within this range, the LCD displays a dispensing message and both servos rotate from 0° to 90°. The tap remains actuated for approximately three seconds. The servos then return to 0°, and the LCD displays "Thank You" and "Visit Again" before the system resumes detection.

System Flow



5. Hardware Components

Component	Purpose
Arduino Uno	Main controller
HC-SR04	Non-contact distance/hand detection
16×2 I2C LCD	Status and distance display
2 × servo motors	Tap/mechanical actuation
18650 battery/power system	Portable power, subject to final implementation
3D-printed enclosure	Mounting and protection

6. Circuit Connections and Technical Settings

Item	Connection / Setting
HC-SR04 TRIG	D9
HC-SR04 ECHO	D10
Servo 1 signal	D6
Servo 2 signal	D7
I2C LCD SDA	A4
I2C LCD SCL	A5
LCD I2C address	0x27
Detection threshold	15 cm
Open / closed angle	90° / 0°
Dispensing duration	3 seconds

Figure placeholder: Add the final verified circuit diagram in the website/final submission version.

7. Software and Arduino Programming

The Arduino program uses `Wire.h` for I2C communication, `LiquidCrystal_I2C.h` for the display and `Servo.h` for the two actuators. It initializes all hardware, calculates ultrasonic distance from echo pulse duration, displays the measured distance, checks the 15 cm threshold and executes the timed dispensing sequence. The supplied source code is suitable for documenting Phase 1. Future revisions can add an ultrasonic timeout, non-blocking timing and additional safety/state logic.

8. 3D CAD and Mechanical Design

Phase 1 includes a custom enclosure system for the electronics and servo mechanism. The design provides mounting for the LCD, ultrasonic sensor, Arduino/control electronics, power hardware, wiring and servo actuation. STL files for the main box, lids and servo enclosure can be distributed through the project website. Final CAD screenshots should be inserted into the website and final report when available.

9. Phase 1 Testing and Results

Testing should verify ultrasonic distance measurement, LCD messages, movement of both servos, response at the 15 cm threshold and return to the closed state after the three-second interval. Mechanical testing should confirm that the linkage can operate the intended tap reliably. Quantitative claims about water savings, battery life, dispensing accuracy or long-term reliability should only be made after physical measurements.

10. Current Limitations

The current design uses a fixed detection threshold and a fixed three-second dispensing duration. It does not yet measure water level, flow rate or actual consumption. The program uses blocking delays during the dispensing sequence. Power performance and mechanical durability also require validation on the final physical prototype.

11. Future Scope – Phase 2

Future upgrades can include water-level monitoring, low-water alerts, rechargeable battery charging/protection, ESP32 Wi-Fi connectivity, a mobile/web dashboard, flow measurement, water-usage analytics, configurable dispensing duration, improved mechanical linkage and more robust sensor filtering.

12. Conclusion

Phase 1 integrates ultrasonic sensing, Arduino control, dual-servo actuation, an I2C LCD and custom 3D enclosure development into a clear touchless dispensing architecture. The next milestone is complete hardware assembly and measured testing of the mechanical mechanism, power system and dispensing reliability.

13. References and Website Resources

The final references should include official Arduino documentation, the HC-SR04 module documentation/datasheet, servo specifications, I2C LCD documentation and the CAD software documentation used for enclosure design. The project website can provide the Arduino source code, STL models, circuit diagram and this report as downloadable resources.

Recommended website file structure

```
documents/Automatic_Water_Dispenser_Phase1_Report.pdf
code/automatic-water-dispenser.ino
images/circuit.png
images/cad-*.jpg
models/*.stl
```